



SkillNest.Ik

“Nurturing Skills & Empowering Futures”

Project Team Members_____



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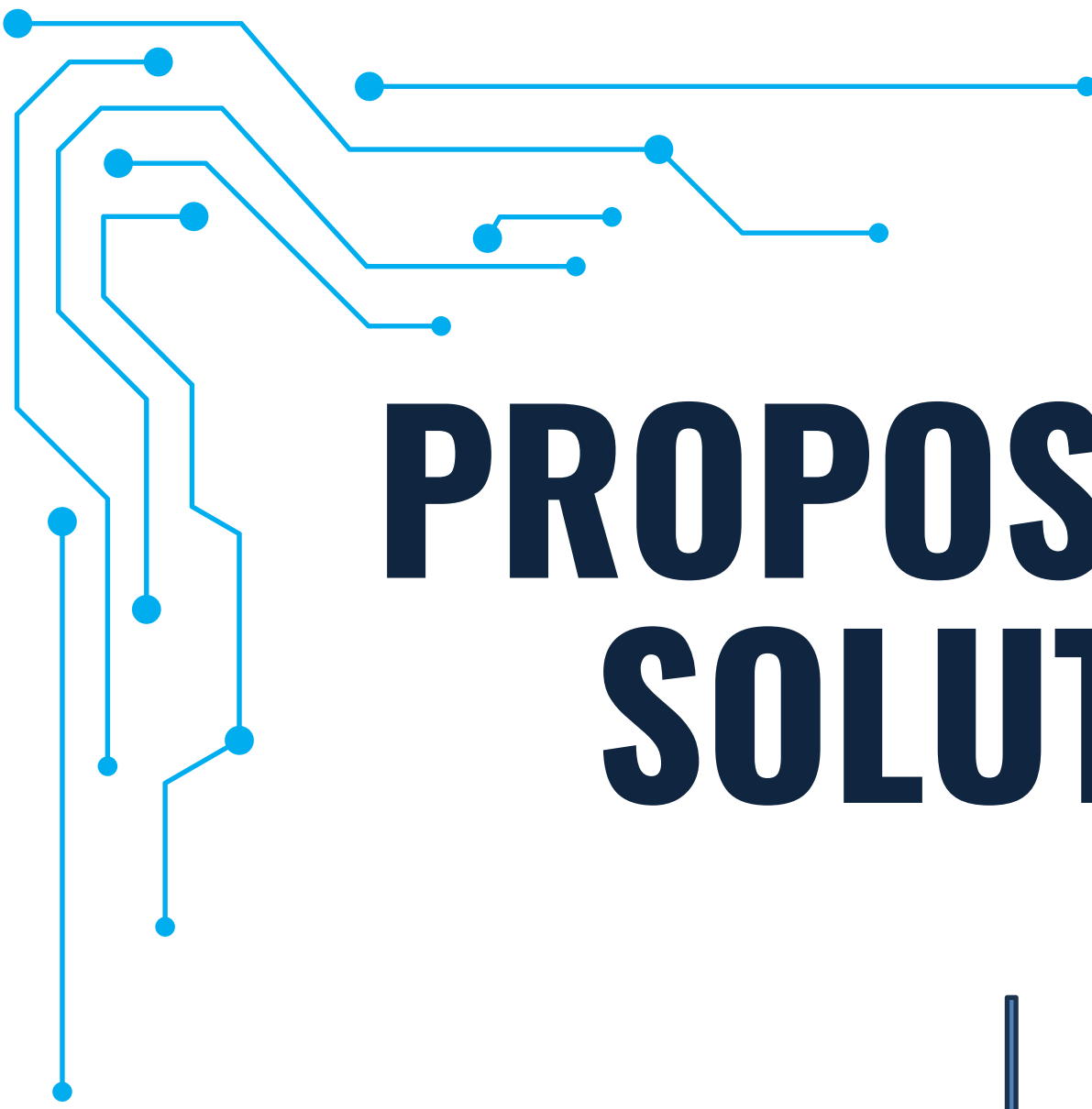


PROBLEM STATEMENT

WHAT PROBLEM ARE WE SOLVING?

- Lack of centralized academic support system
- Difficulty finding suitable peer tutors
- Study materials are scattered and unorganized
- Limited personalized academic guidance





PROPOSED SOLUTION

**A unified
academic
platform
that
provides**

- Secure multi-role user management
- Personalized academic profiles
- Peer tutor matching system
- Quiz & assignment support
- Centralized resource repository
- Smart search & filtering



TECHNOLOGY STACK



MongoDB

NoSQL database for storing users, profiles, resources and submissions.

REST APIs

Communication between frontend and backend services.

Express.js

Backend framework for building REST APIs and handling server logic.

Layered Architecture

Client (React) - API Layer (Express Routes)- Controllers -Database

SYSTEM ARCHITECTURE OVERVIEW

FRONTEND

BACKEND

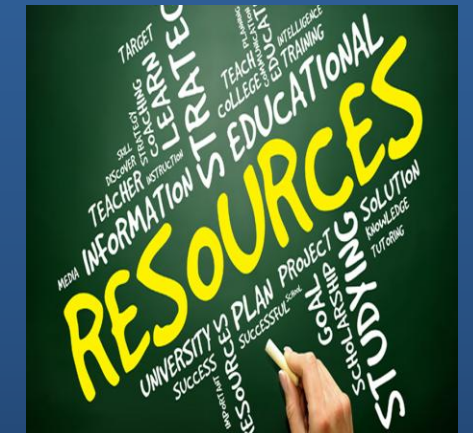
DATABASE



A colorful illustration depicting business growth and teamwork. In the center, a large blue line graph with an upward-pointing arrow is displayed on a black stand. A man in a purple shirt is standing on the left, pointing at the graph. Another man in a blue shirt is sitting on a red box labeled 'BOX' on the right, using a magnifying glass to inspect the graph. A third man in a purple shirt is standing on top of the graph, holding a rocket ship. The background features various business-related icons: a target, a bar chart, a pie chart, and a speech bubble containing a bar chart. The overall scene is set against a light blue background with stylized clouds.



(2) PEER HELP, TUTOR SUPPORT, QUIZ & ASSIGNMENTS HELPING



(3) STUDY RESOURCES & NOTES SHARING

USER MANAGEMENT & ACADEMIC PROFILING_____

Scope & Key Features



- Secure registration, multi-role login and password recovery
- Detailed academic profiles: degree, semester, subjects, skills
- Tracking: strengths, weaknesses and profile completeness metrics

Technical approach: JWT Authentication, Node.js authentication middleware, MongoDB user schema for roles and profiles

HOW THIS MODULE WORKS

- User registers with basic personal details
- System securely encrypts and stores credentials
- User logs in based on assigned role (Student / Tutor / Admin)
- Password recovery available via verification process
- User completes academic profile (degree, semester, subjects, skills)
- System allows users to mark strengths and weak areas
- Profile completeness percentage is calculated automatically
- Profile data is shared with other modules for personalization



PEER HELP, TUTOR SUPPORT & ASSIGNMENT HELPING__

Key Features

- Help request creation
- Tutor matching
- Availability scheduling
- Quiz creation & auto grading
- Assignment submission & feedback
- Tutor credibility rating



HOW THIS MODULE WORK____



- Student submits help request
- System matches suitable tutor
- Tutor accepts request
- Session is conducted
- Student rates the tutor

STUDY RESOURCES AND INTELLIGENT NOTES SHARING MODULE__

Key Feature

- Upload PDFs & links
- Subject & semester categorization
- Search & filtering
- Bookmark & rating system
- Content reporting
- Profile-based prioritization



HOW THIS MODULE WORKS

- Student uploads resource
- System validates file and details
- Admin approval (if required)
- Resource becomes visible in repository
- Users search using subject or semester
- System shows personalized results



SYSTEM INTEGRATION

MODULE INTERCONNECTION

- User profile enables personalization
- Resources support peer help
- Tutors create verified learning content
- Share resources and notes



PROJECT VALUE AND FEASIBILITY

Value to University Students

- One platform for all academic needs
- Personalized learning experience
- Structured peer and tutor support

Feasibility

- MERN Stack based development
- Modular development
- Realistic within semester timeline



The background features a blue-tinted image of the SLIIT (Sri Lanka Institute of Information Technology) building. Overlaid on this are white circuit-like patterns with dots and lines, primarily concentrated on the left and right sides of the image.

THANK YOU !

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